CFDNet: Coupling Computational Fluid Dynamics With Convolutional Neural Networks for Gas Detection Using Thermal Infrared Multispectral Video

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***Abstract*—Gas detection is critically important in both industrial production and environmental monitoring. Due to the absorption characteristics of gases in the infrared wavelength domain, thermal infrared multispectral video imaging provides a convenient data sensing method for rapid, large-scale gas detection. In the process of gas detection, the shape of the detected gas leakage region is often distorted by the irregular motion patterns of gases. Although some studies have considered temporal motion information, these spatial–temporal feature extractors were designed originally for regular and salient objects such as vehicles and pedestrians, but not for gas. Moreover, the problem of weak gas signal and lack of a large well-labeled dataset also hinders the development of gas detection with deep learning. Regarding the above issues, a novel gas detection network, computational fluid dynamics neural network (CFDNet), was proposed for infrared multispectral gas detection. First, to better fit the motion patterns of gas and keep the shape of the detected leakage region, a spatial–temporal fluid motion feature extractor, computational fluid dynamics (CFD) Basic Block, was proposed. CFD Basic Block diffuses and displaces high-dimensional features through a convolution based on the Navier–Stokes equations in CFDs. Second, to enhance the signal of gaseous targets, a local entropy and combination difference (LED) data feature enhancement module was designed based on the instrument characteristics and local entropy information. Finally, a simulated dataset and a transfer learning framework were built to train a deep learning gas detection model with great generalizability. Experiments show that the proposed method, CFDNet, achieves a better performance than existing approaches. On the real-world dataset used for testing, it reaches an IoU of 50.128%, a Kappa of 59.902%, and an $F1$ score of 44.885%. CFDNet demonstrates an excellent performance on keeping the shape of the detected gas leakage region under irregular motion patterns, especially for gas plumes with small-area ratio and low signal-to-noise ratio.**

***Index Terms*—Gas detection, infrared multispectral video imaging, spatial–temporal feature extraction.**

I. Introduction

DETECTING the leakage regions of gases such as carbon dioxide, sulfur dioxide, methane, and ammonia is a significant task for industrial production, environmental protection, and human health [1]. Previous studies indicated that methane is the second most important greenhouse gas contributor [2], and the leakage of toxic gas in the industrial facility may cause terrible casualties. Due to the presence of absorption when electromagnetic waves in the thermal infrared domain pass through gas, the digital number of pixels will be attenuated. This phenomenon was mathematically described by Beer–Lambert's Law [3]. Remote sensing has been applied to large-scale observation and characterization of terrestrial surface features widely [4], [5], [6]. Thermal infrared multispectral remote sensing imaging provides an efficient sensing approach for gas detection [7]. Up to now, quantifying the concentration of gas through chemical reagents [8] is still a widely used method. However, compared to manual inspection, a thermal infrared multispectral camera obtains more data over a large area in a single instance and is safer for the operator. The left side of Fig. 1 illustrates the scene of detecting gas with a thermal infrared multispectral camera.

Infrared imaging has been applied for gas detection since 1996 [9]. The approaches of gas detection can be divided into traditional statistical methods and deep learning methods.

Traditional statistical gas detection methods mainly include moving target detection strategies. Because the motion pattern of gas is continuous, extracting temporal information from video has a great effect on improving performance [10]. The main idea of moving gas target detection lies in establishing a statistical background model based on the first few frames and detecting variational pixels in subsequent frames. The moving target detection algorithms that have been applied to gas detection contain a Gaussian mixture model (GMM) [11], [12], visual background extractor (ViBe) [13], and frame subtraction and integration (FSI) [14], [15]. Traditional statistical methods

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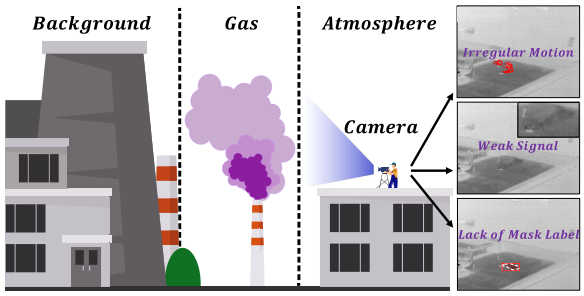


Fig. 1. Left side illustrates the schematic of imaging. The top-right corner displays the problem of irregular motion. The middle-right part displays the problem of a weak signal. The bottom-right part illustrates the lack of a well-labeled mask of the gas leakage region.

have a prerequisite that the camera needs to be fixed, and they are not robust to environmental noise.

With the development of deep learning, significant breakthroughs have been achieved in gas detection with thermal infrared multispectral video imaging, addressing the above challenges. Wang [16] proposed a 3-D CNN-based classification model named VideoGasNet to determine the strength of leakage source across eight classes. Some studies on gaseous target detection focus on extracting the bounding boxes of gaseous targets [17], [18], [19], [20]. Furthermore, semantic segmentation provides an end-to-end strategy to extract the mask of the leakage region, which helps inspectors locate the leakage region of gas more precisely [21]. On the other hand, instance segmentation methods such as MASK R-CNN can obtain bounding boxes and masks of gas at the same time [22], [23].

However, gaseous objects have special characteristics that are greatly different from normal objects in nature images, such as vehicles and pedestrians. Although these issues have attracted the attention of some scholars, they have not been fully addressed, which makes gas detection difficult to achieve.

1) *Irregular Motion Pattern of Gas*: As a typical type of fluid, the motion pattern of gas is easily affected by the wind, making its shape flexible and arbitrary [22], [24], as shown in the top-right corner of Fig. 1. Current spatial–temporal feature extractors applied to moving target detection mainly focus on legible objects that have high saliency and relatively fixed motion trajectories. Although these spatial–temporal feature extractors perform well on natural images, they sometimes fail to keep the shape of the gas leakage region when its shapes change drastically [19]. The Gaussian dispersion model has been deduced as a voxel shift field (VSF) [20] and was applied to gas detection. However, the VSF fails to detect gas when the gas plume is too small.
2) *Weak Signal of Gaseous Target*: In thermal infrared image, gas is semi-transparent because its transmissivity varies from 0 to 1. It means that the signal of the background complex is combined with the signal of the gaseous target. As shown in the middle-right part of Fig. 1, the content present in the gas region is typically similar to that behind the gas. According to the definition of the International Society of Optical Engineering, the local signal-to-noise ratio of weak targets is less than 5 dB [25], while the local signal-to-noise ratio of gaseous targets is often less than 5 dB or even 0 dB. Weak signal of gas makes detecting gas difficult and sensitive to noise [14], [26]. To overcome this problem, multiscale feature extraction has been considered [27]. However, the effectiveness of multiscale feature extraction remains limited.
3) *Deficiency of Large Well-Labeled Video Dataset*: Current datasets for the task of gas detection were collected with discrete frames, lacking temporal information [28]. Moreover, as displayed in the bottom-right part of Fig. 1, labeling precise masks of the leakage region of gas is more difficult, compared with labeling the bounding boxes of the leakage region [29]. In prior work, certain researchers annotated gas leak regions based on expert knowledge [30], while others employed software such as Blender to simulate plumes [23]. Nevertheless, these methods are constrained either by the small scale of the datasets or by the insufficient realism of the simulation.

To solve these challenges, we have proposed a complete gas detection workflow with thermal infrared multispectral video datasets. The key contributions are summarized as follows.

1) A physics knowledge-inspired gaseous targets semantic segmentation model, computational fluid dynamics neural network (CFDNet), was proposed. The physics-inspired modules and deep learning were integrated into a unified workflow. With a physics-inspired block, an entropy-based enhancement module, and a simulated dataset for deep learning, the framework addresses three challenges in thermal infrared multispectral gas detection: irregular motion, weak signals, and lack of labeled data.
2) The computational fluid dynamics basic block (CFD Basic Block) is proposed to predict the displacement of gaseous partials implicitly. This spatial–temporal feature extractor is more suitable for irregular gas plumes and outperforms other encoders, especially in detecting small and dim gaseous masks. To augment the features of gas, a local entropy and combination differentiation (LED)-based feature enhancement module is combined with a spatial–temporal encoder and semantic segmentation decoder. The efficiency of the LED module comes from the spectral characteristics of the multispectral camera and the sensitivity of the gas plume to the local entropy.
3) A simulated video dataset is constructed with the weather research and forecasting model large-eddy simulation (WRF-LES) program. The simulated thermal infrared multispectral video datasets are organized in the form of sequential images. Moreover, a gas detection workflow that combines data simulation and transfer learning is proposed to overcome the semantic gap between the simulated dataset and the real-world dataset.

The remaining sections of this article are structured as follows. Section II provides an overview of research. Section III introduces the dataset. Section IV explains the structure of

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the proposed deep learning network. Section V describes the implementation and the experimental evaluation. Section VI concludes the analysis and discussion. Section VII states the conclusions and future improvements.

II. Related Work

Previous research in the field of gas detection with infrared imaging can be categorized into traditional statistical methods and deep learning methods. In further detail, the deep learning methods can be divided into single-frame models and multi-frame models [29].

A. Statistical Moving Target Detection Methods for Gas Detection With Infrared Imaging

Treating gas as a moving target is the earliest research schema applied in gas detection. These methods separate moving pixels from the static background. The main branch of statistical moving target detection is background modeling [31]. Due to the absorbance characteristics of gas in the thermal infrared region, multiple bands of a thermal infrared image include both reference bands that do not capture absorption and active bands that do. Therefore, reference bands of the image can be tackled as a static background. Thus, the feature of gas can be enhanced by subtraction or division between active and reference bands [32], [33], [34], [35], [36]. On the other hand, the average of first few frames or previous frames also denotes the past status of the scene, which is different from the current frame. The FSI method simply estimates background from previous images and updates it over time [14], [15]. Afterward, some scholars have combined FSI with other manual features such as irregularities and directions to filter environmental disturbs [26], [37], [38], [39], [40], [41], [42]. GMM is a robust background modeling algorithm that was first proposed for natural images [43]. It was also applied to detecting gas with thermal infrared video [11], [12]. Within the scope of GMM theory, background is composed of 3~5 normal distributions with different centers μ_i and standard deviations σ_i. Parameters are initiated with the first few frames that do not contain any gas and updated by subsequent frames with a specific learning rate α. There are some modified approaches based on GMM that were proposed later to exclude false alarms coming from moving humans or vehicles. For example, GMM can be combined with morphology filtering [44], the features from accelerated segment test (FAST) [45], contrast-limited adaptive histogram equalization (CLAHE) [46], and kernel density estimation (KDE) [47]. ViBe is another background model that can be compared to a dictionary, consisting of various words. It is initiated with only the first frame. Therefore, ViBe was used in detecting the leakage region of gas when inspectors have stricter requirements on the speed of responsibility [13], [48].

Besides background modeling methods, optic flow methods can separate pixels with larger motion magnitude [49]. It can be divided into sparse optic flow motion detection method [50] and dense optic flow motion detection method [51]. Sparse optic flow used in gas detection is more precise but more computational [52]. Both background modeling methods and optic flow methods are sensitive to disturbances and imaging noise [53].

B. Single-Frame Segmentation Models for Gas Detection With Infrared Imaging

Detecting gas with classification or object detection approaches mainly focuses on determining the strength of the leakage source or the bounding boxes of the leakage region [16], [17], [18], [19], [20], [54]. However, segmenting the leakage region of gas can extract a precise mask based on machine learning. It helps a lot when inspectors want to assess the extent of gas diffusion. Except for segmentation with handicraft features [55], segmentation models used for gas detection can be divided into instance segmentation and semantic segmentation.

Instance segmentation models mainly include two-stage methods, such as Mask R-CNN [56], and single-stage methods such as Mask DINO [57] and Mask Frozen-DETR [58]. Although Mask DINO and Mask Frozen-DETR perform well on natural images, the scale of the gas detection dataset is too small to train these Visual Transformer (ViT)-based models [59]. Gu et al. [23] compared the performance of different CNN-based instance segmentation models on their simulated gas segmentation dataset. Mask R-CNN reaches the state of the art among CNN-based instance segmentation methods, including Mask R-CNN, MS R-CNN [60], discriminative loss [61], harmonic [62], and spatial embeddings [63] in the task of gas detection. Moreover, an edge attention mechanism was added to Mask R-CNN to refine the boundary [22].

As to semantic segmentation, it plays a more important role on gas detection because discriminating different gas plume objects from each other is not necessary on some specific occasions. UNet [64] has been widely applied in methane detection on satellites remotely [2], [21], [65]. It gains better results than traditional statistical methods. Afterward, a new multiscale semantic segmentation model named PUNet [27], based on PSPNet and UNet, was presented. Considering that infrared imaging is vague and simple, Wang [30] continually fused multimodal data to segment gas, including thermal infrared images and visible natural images. The proposed model, called RT-CAN, outperforms the segmentation model with single-modal inputs or other multimodal fusion strategies dealing with gaseous targets.

C. Multiple-Frame Spatial–Temporal Feature Extraction Neural Networks

Just as mentioned above, adding temporal information into the detection model is beneficial [10] because the diffusion of gas is a continual procedure affected by wind, temperature, and so on. Spatial–temporal feature extraction used in action recognition has made great advancements in the past few years. To extract the motion pattern of objects, a two-stream network [66] was first introduced to deal with visible images and optic flow images at the same time. Because calculating optic flow is time-consuming, 3-D CNN processes multiple frames at the same time and demonstrates good performance at action recognition [67]. Bhatt et al. [24] combined the UNet semantic segmentation framework and 3-D CNN to segment the diffusion field of vapor, forming a method named ST-UNet. Afterward, a lighter spatial–temporal feature extractor based on R(2+1)D CNN appeared to increase the speed of inference

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later [68]. It separates 3-D CNN into spatial 2-D CNN and temporal 1-D CNN and has been proven to have remarkable performance. Similarly, Lin [69] proposed 2.5D-UNet, which also uses separated temporal convolution to extract motion information. In addition to gas detection, temporal modeling methods based on 2-D CNNs, dynamic temporal modeling has also attracted extensive attention in other fields. For example, memory mechanisms [70], adversarial neural networks (ANNs) [71], Transformer-based video understanding [72], and large language model (LLM)-based video understanding [73] have all been proposed in recent years. However, in the field of gas detection that we are studying, the performance of these state-of-the-art video understanding methods on small-scale gas leak detection datasets remains limited due to the constraints of dataset size. Therefore, 2-D CNNs, which are less demanding in terms of data scale, remain the mainstream models for gas detection tasks [23].

Some other 2-D CNN-based spatial–temporal feature extractors, aside from those involving modifications to temporal convolution, have been presented. ConvLSTM integrates long short-term memory (LSTM) and CNN together [74]. Temporal shift module (TSM) is a light and efficient video understanding network that shifts features along the dimension of time [75]. TSM evolved to the temporal excitation and aggregation (TEA) model afterward, containing an additional motion excitation (ME) module [76]. Motion squeeze network (MSNet) is a neural motion feature learning model for video understanding, which combines optic flow estimation and convolution together [77]. Temporal difference networks (TDNs) include an efficient temporal module (TDM) that explicitly leverages a temporal difference operator and systematically assesses its impact on both short- and long-term motion modeling [78]. They have been applied and compared on the insubstantial object detection video (IOD-Video) dataset later [19], and a new gas detection-oriented loss function, called spatiotemporal aggregation loss (STAloss), was designed by Zhou. Although these models are effective in determining the bounding boxes of gas, extracting the masks of the leakage region is difficult for them because of the irregular motion pattern and arbitrary shape. Therefore, a novel gas detection model, the VSF model, was proposed by Zhou et al. [20], inspired by the Gaussian dispersion model. The VSF outperforms traditional spatial–temporal feature extractors but still fails with small gas plumes.

III. Datasets

Besides the IOD-Video dataset used for pretraining, our dataset includes a real-world infrared multispectral video dataset and a simulated infrared multispectral video dataset.

A. IOD-Video Wideband Infrared Video Dataset

Insubstantial object detection (IOD)-Video dataset was constructed by Zhou et al. [19] and published in 2022. They have recorded the emission of smoke, water vapor, and VOC gas leakage (e.g., olefins, alkanes, and carbon monoxide) with 600 videos, up to 141 017 frames. Because the images in IOD-Video are grayscale, there are only three bands in every infrared image. In addition, the labels of bounding boxes are included in this dataset. Table I summarizes the main information of IOD-Video.

TABLE I
Information of IOD-Video

	Videos	Frames	Gases	Channels	Height	Width
IOD-Video	600	141,017	5	3	480	640

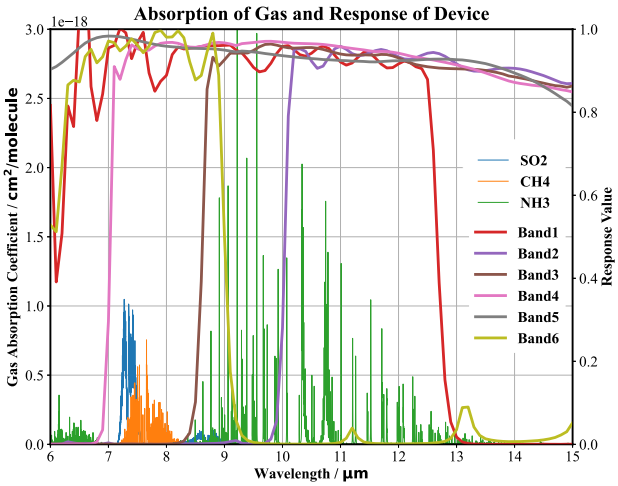


Fig. 2. HITRAN absorption coefficients of SO_2, CH_4, and NH_3 [79], as well as the response curve of thermal infrared multispectral camera [7].

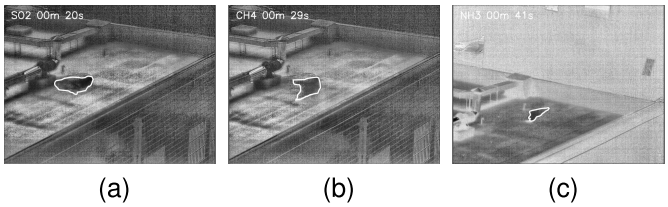


Fig. 3. Scenario of real-world datasets. (a) SO_2. (b) CH_4. (c) NH_3.

B. Real-World IR MS Video Dataset

In order to explore the temporal routine of gas emission with thermal infrared multispectral imaging, thermal infrared multispectral video was collected by an uncooled six-band thermal infrared camera. The IFOV of the camera is $12° \times 10°$ and the noise equivalent temperature difference (NETD) of the camera is 50 mK @ 25 °C/F#1, where F# = 1 [7]. The working wavelength of the imaging optical system is 6.5 ~ 15 μm. Real-world videos recorded the emission scenes of sulfur dioxide (SO_2), methane (CH_4), and ammonia (NH_3) under control. Fig. 2 displays the absorption coefficients of three types of gas and the response curve of the used multispectral camera.

With the gas emitting, the camera recorded the scene once per second. The size of every image is 480 × 640 pixels. As to the real-world dataset, 3 videos with 260 frames were captured. Fig. 3 displays the scenario of real-world datasets.

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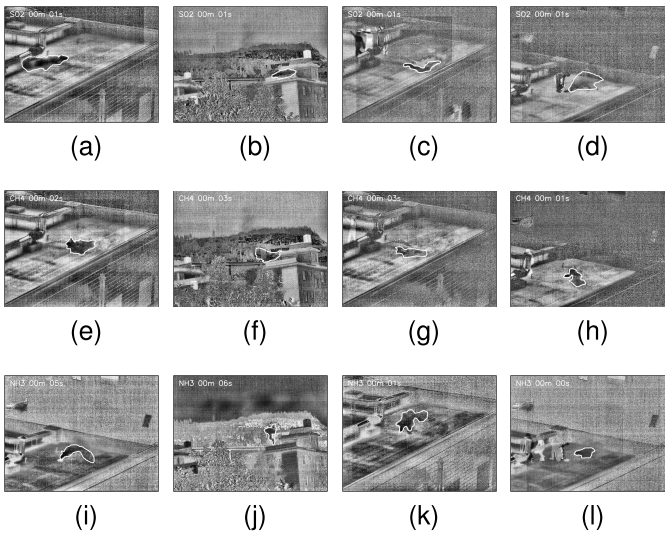


Fig. 4. Thumbnails of 12 scenarios in simulated datasets. The snapshots of the video are displayed as (a)–(l).

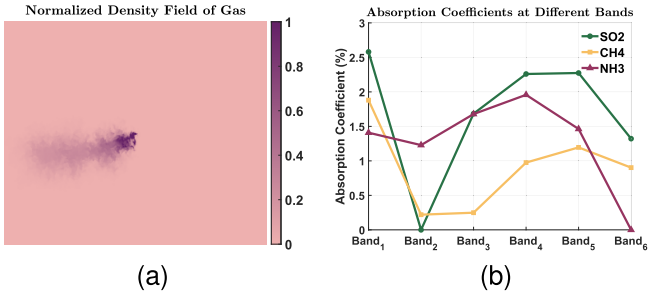


Fig. 5. Data simulation. (a) Demonstration of normalized SO_2 density field. (b) Absorption of gas in standard atmosphere, where $T = 300$ K and $P = 1$ atm.

C. Simulated IR MS Video Dataset

To enlarge the thermal infrared multispectral video dataset for training a deep learning model and generate labels automatically, we constructed a simulated dataset with the WRF-LES. The WRF-LES model is a state-of-the-art mesoscale numerical weather prediction system designed for both atmospheric research and operational forecasting applications. It was used for the numerical simulation of the gaseous concentration field. Fig. 4 presents the scenario of simulated datasets. The bands and sizes of simulated images are the same as real-world images. To simulate images as real as possible, we not only selected static backgrounds but also picked consecutive frames to form the dynamic backgrounds. According to Beer–Lambert's law [3], the image I received by the sensor of the camera has been weakened by the absorption of the gas plume, which can be expressed as follows:

$$I \approx I_{bkg}\left(1 - \gamma_{\text{gas}} \cdot \text{cl}\right) = I_{bkg}\left(1 - \gamma_{\text{gas}} \cdot L \cdot \rho\right) \tag{1}$$

where γ_{gas} is the absorption coefficients of gas, cl denotes the column concentration of gas, L is the optic path length, and ρ represents the density of gas. In the simulation procedure, we select some images without gas plumes to get the background I_{bkg}, while the density field of the gas ρ is determined by WRF-LES [80]. The absorption coefficients of related gas γ_{gas} are determined by HITRAN [79] compiled parameters database and the response curve of the camera provided by the manufacturer. Fig. 5(a) displays the normalized density field of the gas obtained by simulation. Fig. 5(b) presents the convolved response curves of the related gas on the used thermal infrared multispectral camera.

In summary, we have simulated the emission of three types of gases in 12 kinds of environments. The density field of gas is calculated by random situations and CFD. Finally, up to 12 videos with 2400 frames were simulated with the FORTRAN program. The information of the dataset we proposed can be summarized in Table II.

TABLE II
INFORMATION OF CONSTRUCTED DATASETS

	Videos	Frames	Gases	Channels	Height	Width
Real-world	3	260	3	6	480	640
Simulated	12	2400	3	6	480	640

IV. Methodology

The proposed CFDNet model is based on ResNet [81], which is a classic convolutional backbone used in various computer vision downstream tasks. This end-to-end model processes thermal infrared six-channel videos and outputs the mask of gas leakage region for each frame. CFDNet is composed of three parts. The first component of CFDNet is a new data enhancement block designed for infrared multispectral images. The second component is a convolutional temporal spatial–temporal encoder, which aims to extract the high-level feature of data along the dimension of space and time, and the third one is a semantic segmentation decoder.

As shown in Fig. 6, the innovative modules proposed are highlighted in blue and orange colors. Their concise illustrations are placed at the bottom of the structural graph of structure. CFD-based spatial–temporal encoder can be treated as a universal baseline for understanding the video of fluid motion. Adding physical laws to deep learning neural networks is an effective strategy for improving the performance and generalization of the network.

A. LED Data Enhancement Module

Remote sensing images often suffer from degradation, noise, and pixel-level variations during imaging, especially for thermal infrared images [82], [83]. The contrast of the single wideband thermal infrared image is normally lower than natural visible image. However, compared with a single wideband thermal image, a thermal infrared multispectral image has additional spectral information. Moreover, local entropy has a great effect on increasing the saliency of gaseous targets. On these bases, we concatenate the combination of difference and local entropy of the infrared image, proposing a local entropy and combination of difference-based (LED) data enhancement module.

Denoting the input video as $V \in \mathbb{R}^{B\times N\times T\times H\times W}$, each frame in batch can be presented as $I \in \mathbb{R}^{N\times H\times W}$, where B is the size of batches, N is the number of bands or channels,

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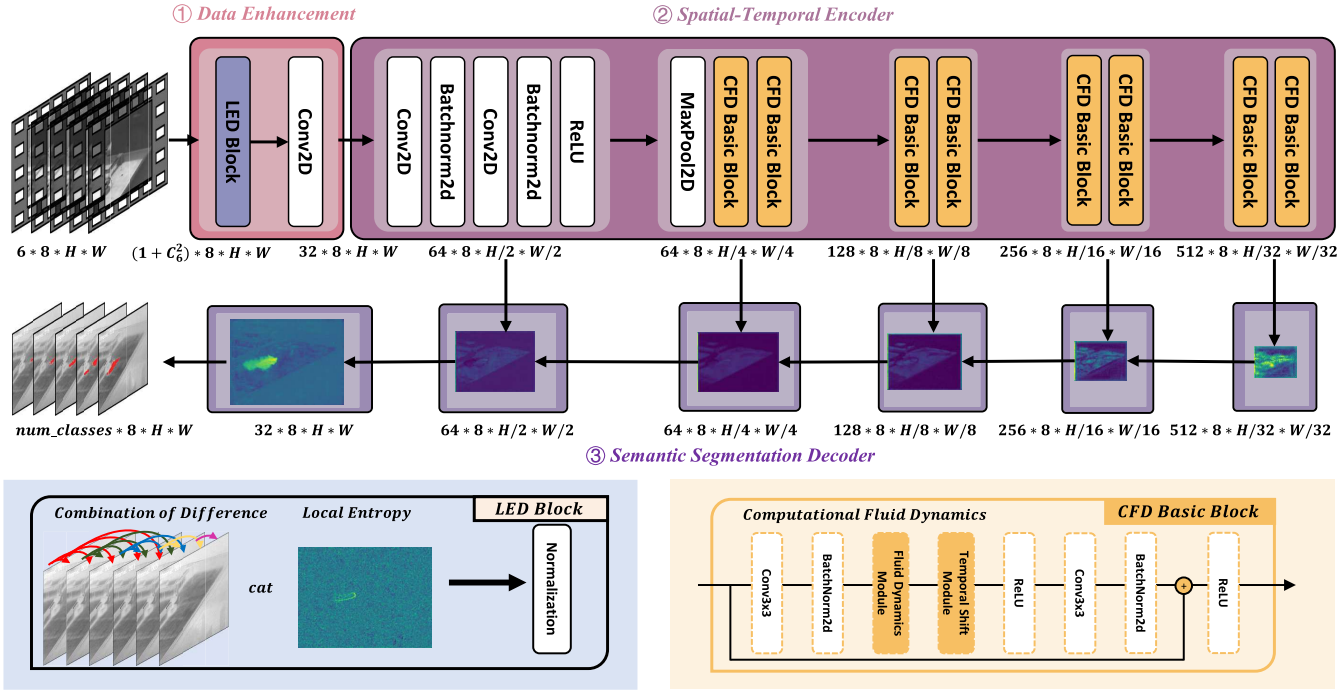


Fig. 6. Overall process of the proposed approach. Based on data enhancement and a convolutional neural network, we deduce the Navier–Stokes function of CFDs. The model first enhances the feature of gas by a combination of differentiation and local entropy. Then, the low-level features of sequent frames are processed by a spatial–temporal encoder. CFD Basic Block and fluid motion squeeze module predict the diffusion and advection of feature F at time t and position (i, j). Finally, the masks of the leakage region are output from the general UNet semantic segmentation decoder.

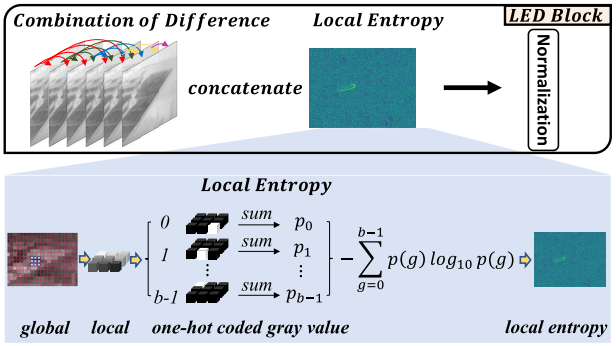


Fig. 7. Details of local entropy and combination of difference-based data enhancement module. LED mainly concatenates the combinatorial difference between each two bands and the local entropy of a specific band.

T is temporal length, H is the height of frame, and W is the width of frame. For the thermal infrared multispectral video input, $N = 6$ in this case. Because the absorption of gas varies along wavelength in the thermal infrared domain, the gray value in the gas leakage region is different between different bands. Subtracting the strong-absorption band and the weak-absorption band removes as much background information as possible. As shown in the upper left part of Fig. 7, the combination of differentiation of I can be presented as $D \in \mathbb{R}^{C_6^2 \times H \times W}$, namely, the concatenation of difference between each two bands

$$D = \text{cat}\left[I_i - I_j \middle| 1 \le i < j \le 6\right] \tag{2}$$

where I_i means the ith band of I. Because I has six bands, the number of channels of D is the binomial coefficient of 6, namely, $C_6^2 = 15$.

After obtaining the combination of differentiations, a specific channel of D is selected to calculate the local entropy of frame I. We have compared the local standard deviation, the first-order gradient, the second-order gradient, and the local entropy of the thermal infrared image that records the scene of gaseous leakage. The local entropy has the most obvious advancement in enhancing the feature of gaseous targets. Since the 8th band of D has the most salient feature for this camera, especially, we select D_8 as D_s to calculate local entropy E.

Here, we introduce a new approach to obtain local entropy more effectively. Compared with the traditional method that statistics the histogram of every local region, this approach calculates local entropy without circulation. First, the image $D_s \in \mathbb{R}^{H \times W}$ is reclassified into integer image $D'_s \in \mathbb{Z}_b^{H \times W}$ with b integral gray scale levels

$$D'_s = \left\lfloor \frac{D_s - \min(D_s)}{\max(D_s) - \min(D_s)} \cdot b \right\rfloor. \tag{3}$$

Then, D'_s is coded by one-hot coding, and we can get a binarized tensor $B_s \in \mathbb{Z}_2^{b \times H \times W}$. If the gray value of D'_s at position (i, j) is b, $B_s(b, i, j) = 1$

$$B_s(b, i, j) = \begin{cases} 1, & D'_s(i, j) = b \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Thenumber of pixels at the gth gray scale level in the local 7×7 region is obtained by a convolutional sum

$$p_g(i, j) = \sum_{m=i-3}^{i+3} \sum_{n=j-3}^{j+3} B_s(g, m, n). \tag{5}$$

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The local entropy E is calculated by p_g

$$E = -\sum_{g=0}^{b-1} p_g \log_{10} p_g. \tag{6}$$

As shown at the bottom of Fig. 7, this procedure can calculate local entropy with a convolutional operation, which is faster and makes the whole model end-to-end.

Finally, input images are converted into a tensor of feature $F \in \mathbb{R}^{B\times C\times T\times H\times W}$ in high-dimensional space, where $C = 16$. To fully utilize the advancements of combinatorial differentiation and local entropy, the high-dimensional feature F of input frame I is presented by the concatenation of D and E

$$F = \text{cat}\,[D, E]\,. \tag{7}$$

B. Computational Fluid Dynamics

As mentioned above, the motion pattern of gas is irregular and arbitrary, which causes the distortion of the detected leakage region under complex conditions. General spatial–temporal encoders were designed for salient solid objects, such as vehicles and pedestrians, whose routines are regular. The displacement of gaseous particles follows the law of CFDs. Therefore, it is natural to consider predicting the routine of gaseous particles under physical laws. Inspired by CFDs, we present an innovative spatial–temporal encoder based on CFDs. CFD is a branch of fluid mechanics that uses numerical analysis and data structures to analyze and solve problems that involve fluid flows. Before introducing the principle of the proposed spatial–temporal feature extractor on enhanced temporal features, it is essential to explain the basic principles of CFD here. The fundamental basis of almost all CFD problems is the Navier–Stokes equations [84], which define many single-phase (gas or liquid, but not both) fluid flows. These equations, however, only admit analytical solutions in very simple cases. With the development of computers, numerical algorithms have been created to estimate an approximate solution of these equations. In the 2-D realm, Navier–Stokes mathematical equations contain the function of the gas velocity field and the function of the gas density field

$$\frac{\partial \mathbf{u}}{\partial t} = -(\mathbf{u}\cdot\nabla)\,\mathbf{u} + v\nabla^2\mathbf{u} + f \tag{8a}$$

$$\frac{\partial \rho}{\partial t} = -(\mathbf{u}\cdot\nabla)\,\rho + \alpha\nabla^2\rho + S \tag{8b}$$

where $\mathbf{u}$ denotes the velocity field, v is the viscosity of gas, f is the force applied outside, ρ is the density field of gas, α is the diffusion of gas, and S is the emitting source of gas.

Equation (8a) explains the change rate of velocity $\mathbf{u}$ with respect to time t. It is composed of three terms. The first term, $-(\mathbf{u}\cdot\nabla)\mathbf{u}$, presents the advection of velocity field. The second viscous item, $v\nabla^2\mathbf{u}$, describes the internal friction within the fluid due to its viscosity v, and the last one f is the stress applied on gaseous particles. Equation (8b) describes the change rate of gas density over time. The first term, $-(\mathbf{u}\cdot\nabla)\rho$, is the advection of gas density that describes the transportation of gas density due to velocity. The second part, $\alpha\nabla^2\rho$, represents diffusion, modeling how ρ spreads out due to diffusion. The third item is the source of gas. We mainly focus on the density

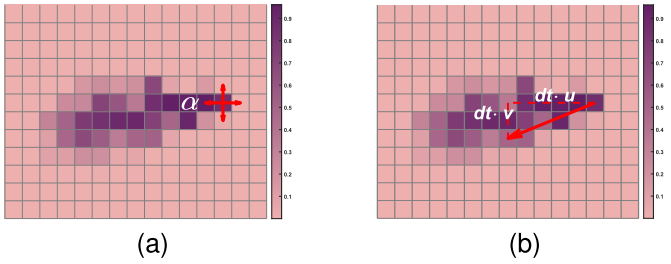


Fig. 8. Motion pattern of gas density field. (a) Diffusion. (b) Advection.

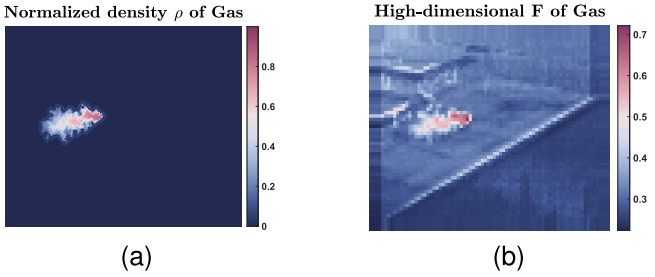


Fig. 9. Comparison between the density of gas and the high-dimensional feature. (a) Density of gas. (b) Visualization of a high-dimensional feature. We selected a specific batch and performed averaging over the channel dimension for visualization. The blurring of the boundaries in a high-dimensional feature map arises from the absence of boundary conditions in convolution.

(8b) here, and the motion pattern of gas can be deduced as two individual processes, namely, diffusion and advection.

Discretizing the 2-D space into a grid, we can idealize the gas density at each position (i, j) as a particle $\rho(i, j)$. As shown in Fig. 8(a), diffusion of gas density represents the phenomenon that the gas at a given point exchanges with its surroundings in space

$$\rho_t\,(i, j) = \rho_{t-1}\,(i, j) + \alpha\nabla^2\rho_{t-1}\,(i, j) \tag{9}$$

where ρ_t and ρ_{t-1} record the status of gas density at different time, t and $t-1$. Equation (9) adopts the form of forward differentiation instead of backward differentiation to reduce computational cost. The difference between forward differentiation and backward differentiation can be ignored when the interval of time dt is small enough. Fig. 8(b) illustrates the advection of gas, which describes the transportation of gas caused by the velocity field $\mathbf{u} = (u, v)$

$$\rho_t\,(i, j) = \rho_{t-1}\,(i + \text{dt}\cdot u, j + \text{dt}\cdot v) \tag{10}$$

which is the numerical expression of equation $-(\mathbf{u}\cdot\nabla)\rho$. $(\text{dt}\cdot u, \text{dt}\cdot v)$ presents the displacement of a gaseous particle from the last time $t-1$ to the current time t. By discretizing the concentration field equation in the Navier–Stokes equations, we can iteratively and numerically compute the state of the gas density field at a fixed time interval dt. Thus, the irregular motion pattern of gas can be described with the deduced CFDs.

C. CFD-Based Spatial–Temporal Encoder

According to Beer–Lambert's law in (1), the relationship between the density of gas ρ and the captured image I is approximately linear. Moreover, the multiplier item $I_{bkg}\cdot\gamma_{\text{gas}}\cdot L$ can be suppressed by the linear fitting of the convolution

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layer in the process of training. As shown in Fig. 9, after the operation of enhancement and feature extraction, the high-dimensional feature F is consistent with ρ.

Therefore, applying the same motion pattern for the high-dimensional feature is feasible and rational for spatial–temporal feature extraction. This replacement can be described as follows:

$$\rho \rightarrow F_{\mathrm{kb}} \tag{11}$$

where $F_{\mathrm{kb}} \in \mathbb{R}^{T\times H\times W}$ denotes the feature map of kth channel and bth batch. Furthermore, $F_{\mathrm{kb}}(t,i,j)$ presents the value of the feature at time t and position (i,j). The main components in CFD are the diffusion item and the advection item, which could be implemented by shifting and the Laplace operator. Thus, the motion pattern of fluid can be extracted with a convolutional module, namely, CFD Basic Block.

Here, we adopt ResNet as a backbone to extract the spatial features of an image using multiple convolutional neural layers and residual connections. The whole encoder is composed of four CFD Basic Blocks, and fluid ME modules mainly predict the field of diffusion of velocity with convolutional operations implicitly. Considering implementing (9) with feature F_{kb}, we imitate the action of diffusion by Laplace operator

$$F_{\mathrm{kb}}(t,i,j) = F_{\mathrm{kb}}(t-1,i,j) + \alpha \cdot \nabla^2 F_{\mathrm{kb}}(t-1,i,j) \tag{12}$$

where the second-order gradient of the feature is

$$\begin{aligned}&\nabla^2 F_{\mathrm{kb}}(t-1,i,j)\\ &= F_{\mathrm{kb}}(t-1,i-1,j) + F_{\mathrm{kb}}(t-1,i+1,j)\\ &\quad + F_{\mathrm{kb}}(t-1,i,j-1) + F_{\mathrm{kb}}(t-1,i,j+1)\\ &\quad - 4F_{\mathrm{kb}}(t-1,i,j).\end{aligned} \tag{13}$$

In convolutional neural networks, the second-order gradient of a feature can be represented by the convolution of the feature and the Laplace operator. Advection describes the displacement of gaseous particles, caused by velocity $\mathbf{u} = (u,v)$. Therefore, we argue that the advection of gas as shown in (10) can be presented as the displacement of a feature across a fixed time interval dt

$$F_{\mathrm{kb}}(t,i,j) = F_{\mathrm{kb}}(t-1, i+\mathrm{dt}\cdot u, j+\mathrm{dt}\cdot v) \tag{14}$$

which means that the feature $F_{\mathrm{kb}}(t-1,i,j)$ at time $t-1$ and position(i,j) moves to $(i+\mathrm{dt}\cdot u, j+\mathrm{dt}\cdot v)$ at time t.

In (12) and (14), parameters α and $\mathbf{u} = (u,v)$ need to be solved. As shown in Fig. 10, to estimate $\alpha \in \mathbb{R}^{1\times H\times W}$ and $\mathbf{u} \in \mathbb{R}^{2\times H\times W}$, convolution layers (Conv2d) and batchnorm layers (BN2d) are used to get these parameters

$$\alpha = \sigma(\mathrm{BN2}d(\mathrm{Conv2}d(F_{\mathrm{kb}}(t-1),\Theta_\alpha),\zeta_\alpha)) \tag{15a}$$

$$\mathbf{u} = \sigma(\mathrm{BN2}d(\mathrm{Conv2}d(F_{\mathrm{kb}}(t-1),\Theta_{\mathbf{u}}),\zeta_{\mathbf{u}})) \tag{15b}$$

where σ is the sigmoid activation function and $\mathbf{u}$ can be split as the horizontal component of velocity $u \in \mathbb{R}^{1\times H\times W}$ and the vertical component of velocity $v \in \mathbb{R}^{1\times H\times W}$ at the dimension of channel. After flow occurs, the changed feature F'_{kb} was fused with the original feature F_{kb} with a custom feature fusion operation as shown in the following equation:

$$F'_{\mathrm{kb}} @ F_{\mathrm{kb}} = F'_{\mathrm{kb}} + F_{\mathrm{kb}} \cdot \sigma\left(\mathrm{AvgPool2}d\left(F'_{\mathrm{kb}} - F_{\mathrm{kb}}\right)\right). \tag{16}$$

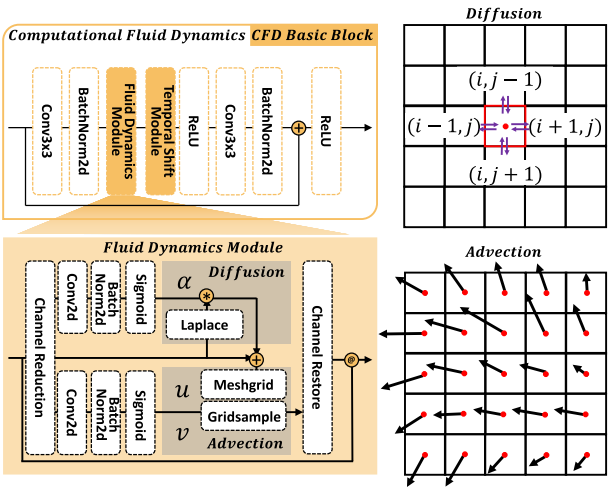


Fig. 10. Diagram of CFD Basic Block. CFD Basic Block is the combination of the FDM, TSM, and the basic block in ResNet.

Following the fluid dynamics module (FDM), the TSM is added into the CFD Basic Block in ResNet [75] to fit the pattern of feature movement along a longer temporal range adaptively. The motion relationship between two adjacent video frames is modeled by mimicking the Navier–Stokes equations in the FDM. However, considering only the previous or the next frame is insufficient. Therefore, TSM is incorporated to enable the FDM to establish connections across multiple frames

$$F_{\mathrm{kb}}(t) = \begin{cases} F_{\mathrm{kb}}(t-1), & 0 \le k < C/8\\ F_{\mathrm{kb}}(t+1), & C/8 \le k < C/4\\ F_{\mathrm{kb}}(t), & C/4 \le k < C. \end{cases} \tag{17}$$

When initializing TSM, we shift 1/8 of the channel features forward by one frame and another 1/8 backward by one frame, while keeping the remaining 3/4 of the channel features unchanged, as shown in (17). The initial shift ratio follows the design of the TDN model [78]. Meanwhile, the temporal shift parameters in TSM are learnable. During subsequent training, the model can adaptively perform weighted fusion of the outputs from the proposed FDM across three adjacent frames, thereby enabling spatial–temporal feature extraction over a longer temporal range.

D. Semantic Segmentation Decoder

The UNet semantic segmentation decoder [64] is responsible for upsampling the feature maps to reconstruct the segmentation map. It uses upconvolution layers to increase spatial resolution and applies skip connections from the encoder to preserve fine-grained details. These connections help improve segmentation accuracy, especially at object boundaries. The final layer generates a pixel-wise classification map, making UNet particularly effective for tasks like gas leakage region semantic segmentation.

V. Experiments and Results

A. Implementation Details and Experimental Settings

Due to the gap in distribution between the simulated dataset and real-world datasets, we adopt the strategy of transfer

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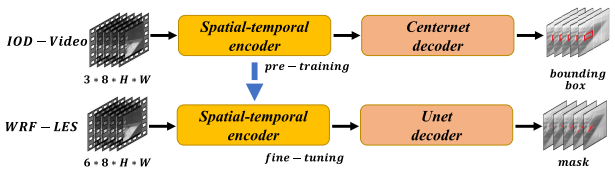


Fig. 11. Diagram of transfer learning strategy. Gaseous object detection was taken as a pretext task.

learning to overcome this drawback. The spatial–temporal encoder was pretrained by gaseous object detection as a pretext task with IOD-Video dataset [19]. Although the IOD-Video dataset was collected by an infrared single wide camera, the gas plume in real-world scenarios is highly beneficial for extracting spatial gas features with a deep learning model. The number of frames in IOD-Video is 141 017, which is large enough to pretrain a robust encoder. As to the details of the transfer learning strategy, we first train the spatial–temporal encoder with a larger real-world dataset, IOD-Video. Gaseous object detection was accomplished by Centernet, which is an anchor-free target detection model. Fig. 11 illustrates the diagram of transfer learning.

Taking object detection as a pretext task, the pretrained encoder is fully fine-tuned on a simulated infrared multispectral video dataset with WRF-LES. We initialize the encoder with pre-trained weights, and the decoder is trained from random initialization. The channel of input image in IOD-Video is 3, while the band of thermal infrared multispectral image is 6. Therefore, the parameters of pretrained weights in the first layer are repeated to fit the data adopted here. Simulated dataset is used for training and validation, based on WRF-LES, while the real-world dataset is used for testing.

As to the experimental settings of traditional statistical methods, the first few frames without the appearance of humans or other disruptions are selected to train the background model. The proposed model, CFDNet, and other deep learning methods were trained on an NVIDIA TITAN RTX GPU with 24 GB of VRAM. We select Adam as the optimizer and initialize the learning rate as 0.0001 with a decay rate of 0.5. If the loss value on the validation dataset does not decrease for four epochs, the learning rate will be reduced by the given decay rate; 70% of the simulated dataset is selected for training, while 30% of the simulated dataset is used for validation. Every multiframe deep learning model was trained for 100 epochs with a batch size of 2. The number of frames that are input into CFDNet is set to 8, which means that 16 frames of thermal infrared images are input into the spatial–temporal feature extraction encoder. The training batch size of single-frame models is set to 16. The simulated videos were sampled at a frame rate of 25 fps, while the real-world videos were captured at 1 fps. To ensure that the trained model can adapt to different frame rates, a dynamic sampling strategy [85] was applied to the simulated training set. Specifically, the original simulated videos were resampled to 25, 5, 2.5, and 1 fps, and all temporal sequences were organized into clips of eight consecutive frames. For the real-world testing dataset, eight consecutive frames of thermal infrared multispectral images were sampled at the original frame rate of 1 fps and then fed into the model. Since the real-world dataset is hardware-limited to a maximum of 1 fps, each of the eight frames exhibits considerable variation, allowing them to be treated as key frames. The dynamic sampling strategy used in training improves the model's inference performance on low-frame-rate cameras.

To supervise these models with ground truth, binary cross entropy (BCE) loss (18a) and Dice similarity coefficient loss (18b) are summed as (18c) to minimize the discrepancy between the prediction output p and the label y

$$\text{Loss}_{\text{BCE}}(p,y) = -\left(y\log(p) + (1-y)\log(1-p)\right) \tag{18a}$$

$$\text{Loss}_{\text{Dice}}(p,y) = 1 - \frac{2\cdot|p\cap y|}{|p|+|y|} \tag{18b}$$

$$\text{Loss}(p,y) = \text{Loss}_{\text{BCE}}(p,y) + \text{Loss}_{\text{Dice}}(p,y). \tag{18c}$$

Five widely used metrics were also deployed to measure the performance of the model, namely, intersection over union (IoU), Kappa, $F1$ score, precision, and recall. The range of values for Kappa is from -1 to 1, while the range of values for other metrics is from 0 to 1. All of these metrics indicate better performance as they approach 1. Among them, IoU, Kappa, and $F1$ score are relatively more comprehensive evaluation metrics that consider class imbalance

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{19}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{20}$$

$$F1\ \text{score} = \frac{2\cdot\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}} \tag{21}$$

$$\text{Kappa} = \frac{P_o - P_e}{1-P_e} \tag{22}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{23}$$

where

$$P_o = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{TN}+\text{FN}} \tag{24}$$

$$P_e = \frac{(\text{TP}+\text{FP})(\text{TP}+\text{FN})+(\text{TN}+\text{FP})(\text{TN}+\text{FN})}{(\text{TP}+\text{FP}+\text{TN}+\text{FN})^2} \tag{25}$$

and TP, TN, FP, and FN represent the true positive, true negative, false positive, and false negative, respectively.

B. Benchmarks

To evaluate the performance of the proposed network, CFDNet, on a constructed thermal infrared multispectral video dataset. We have implemented several methods, including state of the art, on the same backbone and the same dataset. Four kinds of traditional statistical methods (GMM, ViBe, OpticFlow, and FSI), three types of single-frame semantic segmentation approaches (UNet, PUNet, and MaskRCNN), and five kinds of spatial–temporal extractors (ConvLSTM, MSNet, TDNNet, TEANet, and R(2+1)D) were compared in our study.

1) *GMM* [46]: It is a classic statistical moving target detection algorithm, which uses multiple normal distributions to construct the background. It utilizes the first few frames to build a background and detects gaseous pixels in the following frames.

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2) *ViBe* [13]: It is another popular statistical moving target detection method. Its special advantage lies in only using the first frame to build the background. Therefore, ViBe is also an effective and robust approach for gas detection.
3) *OpticFlow* [52]: It mainly estimates the field of optic flow with the assistance of a dense optic flow estimator. Taking gaseous pixels as moving optic partials, OpticFlow can determine the leakage region of gas more accurately, but more time-consuming.
4) *FSI* [15]: It is a simple and fast-moving gaseous target detection approach. The result of FSI can be presented as the Boolean values of consecutive frames.
5) *UNet* [64]: As a classic semantic segmentation model, UNet has been used for gas detection [65]. UNet processes every single frame and outputs the mask of gas without temporal information.
6) *PUNet* [27]: It is made up of the UNet framework and the pyramid scene parsing module. Compared with the common UNet model, PUNet can extract features at multiple scales.
7) *MaskRCNN* [56]: It is a famous instance segmentation model used in computer vision research. It is proved to have magnificent performance on segmenting the leakage region of gas [23].
8) *ConvLSTM* [74]: Long-short temporal memory (LSTM) mechanism was fused with a convolutional neural network to extract the temporal information of gaseous motion. To detect gas with ConvLSTM, we insert LSTM basic cells into the semantic segmentation framework.
9) *MSNet* [77]: It is another video comprehension network, which estimates the opticflow of moving objects with the help of a correlation operation.
10) *TDNNet* [78]: It is an efficient model for enhancing changed information between adjacent frames with differentiation. TDNNet contains short- and long-term temporal difference modules to extract the spatial–temporal features of video.
11) *TEANet* [76]: Deriving from the temporal shift model (TSM), TEANet contains a TEA block. With a ME module and a multiple temporal aggregation (MTA) module, TEANet excels in action recognition tasks.
12) *R(2+1)D* [68]: Due to the high computational complexity of 3DCNN operation, R(2+1)D was proposed to extract temporal information while reducing computational cost. Nevertheless, R(2+1)D is still a time-consuming method compared with other methods.
13) *VSF* [20]: Deduced from the Gaussian dispersion model, the physics-inspired VSF is designed to model geometric irregularities and ever-changing shapes in potential 3-D space.

C. Qualitative Evaluation of Results

Selecting representative image cases to qualitatively assess the performance of different methods, CFDNet has been proven to be a robust gas detection method, which is suitable for many environments.

1) Strong and Salient Signal of Gas: Fig. 12(a) illustrates the scenario of SO_2 emission. Because the absorption feature of SO_2 is relatively evident, most of the gas detection methods have great detection results. Compared with deep learning strategies, traditional statistical methods detect the leakage region of SO_2 with more false alarms. Among traditional statistical methods, GMM and OpticFlow have impressive effectiveness. Although GMM, ViBe, OpticFlow, and FSI can detect most of the leakage region, these methods also detect many pixels of background as foreground. The single-frame gas detection methods, such as UNet, PUNet, and MaskR-CNN, reach higher IoU than statistical methods. However, with the assistance of temporal information, multiframe gas detection methods such as VSF, TDNNet, TEANet, R(2+1)D, and CFDNet have better visualization results in general. The CFDNet reaches the best performance among different methods.

2) Complex Shape Caused by Irregular Motion: Fig. 12(b) records the emission of CH_4. As mentioned in Section I, the distortion of the detected leakage region is caused by the irregular motion pattern of gas. Due to the irregular motion pattern of gas caused by the ever-changing wind field, the shape of the mask where gas exists is complex and arbitrary. In this case, the traditional statistical methods cannot detect any part of the gas leakage region, as there is nearly no true positive (denoted as green) region in the detected results of GMM, ViBe, OpticFlow, and FSI. Specifically, due to the disruption caused by the irregular gaseous motion pattern, single-frame deep learning gas detection methods yield better visualization results than statistical methods, but worse results compared to multiframe deep learning methods. Except for ConvLSTM and MSNet, other multiframe gas detection methods all reach a larger IoU than single-frame gas detection methods, including UNet, PUNet, and MaskRCNN. Among these, the proposed CFDNet demonstrates a great performance in maintaining the boundary information of the detection area, and the IoU of CFDNet reaches the highest.

3) Weak and Small Signal of Gas: Fig. 12(c) and (d) demonstrates the scenario of NH_3 emission. The absorption feature of NH_3 is weak. Therefore, the leakage region of NH_3 is not obvious in the thermal infrared multispectral image. On the other hand, the area of the gaseous target is also very small in these scenarios. In the scene of Fig. 12(c), only TDNNet, TEANet, and CFDNet are able to detect partial gas leakage regions. Among them, TDNNet detects a region significantly smaller than the ground truth, TEANet detects a large number of false positives, and CFDNet achieves the highest IoU of 44.7%. In the scene of Fig. 12(d), only CFDNet achieved relatively better results. Although the single-frame semantic segmentation methods, UNet and PUNet, generally perform well, they sometimes fail to detect small gaseous targets. As the area of the gas plume decreases, the false alarm rate in gas detection increases accordingly. Although PUNet, TDNNet, and R(2+1)D exhibit slightly lower false alarm rates, their missed detection rates are much higher than that of CFDNet. In contrast, CFDNet achieves a relatively low false alarm rate while maintaining a missed detection rate similar to that of TEANet. In summary, compared with other methods, CFDNet has a better performance for weak and small-signal gas detection.

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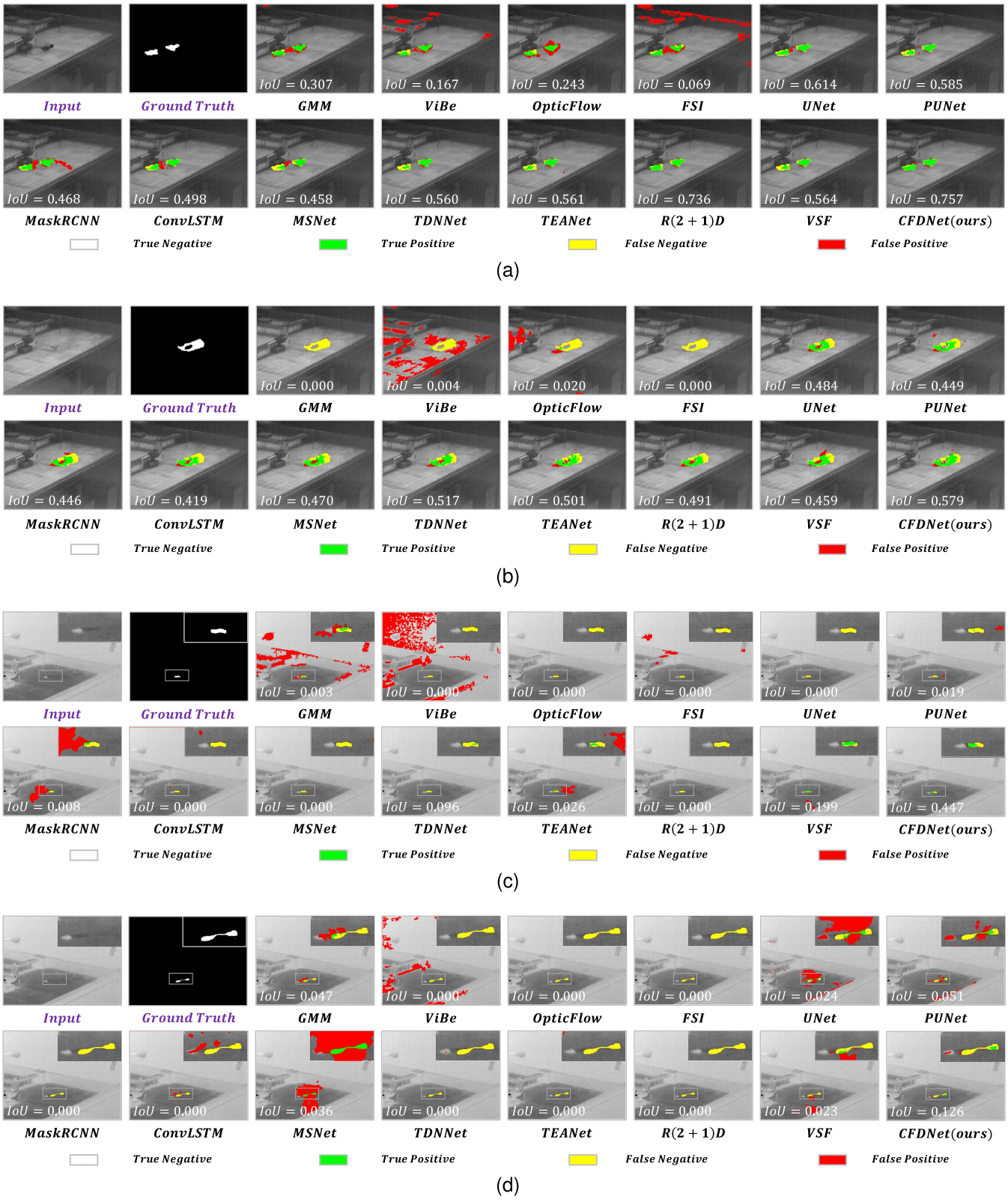


Fig. 12. Typical visualization of results on real-world testing dataset. The GT is composed of both the green and yellow parts. The larger the green portion in the image, the better the performance of the method. The red parts denote false alarms, and fewer of them are better. The yellow parts indicate missed detections, and fewer of them is also better. (a) SO_2 20th frame. This case represents a relatively easy-to-detect scenario because the signal of SO_2 is strong and salient. (b) CH_4 7th frame. This case demonstrates the performance of different methods in detecting smoke plumes with complex and arbitrary shapes. The complex shape comes from the irregular motion pattern of gas in the wind. (c) NH_3 116th frame. This image presents small-area targets due to the low emitting strength. (d) NH_3 132nd frame. These cases also demonstrate the effectiveness of different methods in detecting weak and small targets.

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TABLE III
SCORES OF DIFFERENT METHODS ON REAL-WORLD TESTING DATASET

Types	Methods	Backbone	IoU(%)↑	Kappa(%)↑	F1Score(%)↑	Precision(%)↑	ReCall(%)↑
Statistical Methods	GMM (TIM,2021) [46]	—	16.960	21.394	13.484	14.630	17.067
	ViBe (Photonics,2022) [13]	—	4.847	5.858	4.098	3.293	17.068
	OpticFlow (RS,2020) [52]	—	26.049	30.648	12.407	13.590	16.401
	FSI (JVGR,2018) [15]	—	22.366	27.566	14.725	15.441	20.716
Single-Frame Segmentation	UNet (CVPR,2015) [64]	Resnet18	40.274	49.084	39.990	45.689	40.445
	PUNet (RCAR,2023) [27]	Resnet18	43.680	52.992	43.507	46.239	46.651
	MaskRCNN (ICCV,2017) [56]	Resnet18	42.405	49.542	30.446	33.898	31.840
Spatial-Temporal Segmentation	ConvLSTM (NeurIPS,2015) [74]	Resnet18	21.722	30.440	30.671	28.741	37.463
	MSNet (ECCV,2020) [77]	Resnet18	37.407	46.432	37.500	43.023	38.723
	TDNNet (CVPR,2021) [78]	Resnet18	48.407	57.562	40.650	46.279	41.637
	TEANet (CVPR,2020) [76]	Resnet18	39.542	48.760	41.339	41.043	**49.457**
	R(2+1)D (CVPR,2018) [68]	Resnet18	41.441	48.574	30.895	40.474	29.571
	VSF (TPAMI,2024) [20]	Resnet18	48.008	57.668	43.041	**50.693**	44.448
	proposed	Resnet18	**50.128**	**59.902**	**44.885**	48.553	48.708

D. Quantitative Evaluation of Results

To evaluate different methods more systematically and comprehensively, we statistically analyze five metrics of each method on the testing dataset. The top three scores are marked in red, with the highest score for each corresponding metric bolded.

1) Statistical Methods Versus Deep Learning Methods: As shown in Table III, traditional statistical methods do not perform as well as deep learning methods. Across the five evaluation metrics, the scores of traditional statistical methods are generally lower than those of deep learning methods, whether single-frame segmentation methods or multiframe (spatial–temporal) segmentation methods. For example, the highest IoU, Kappa, and $F1$ scores of statistical methods are lower than 30% in general. The scores of deep learning methods are both larger than those of statistical methods.

2) Semantic Segmentation Versus Instance Segmentation: Dividing the single-frame segmentation methods into single-frame semantic segmentation methods and single-frame instance segmentation methods, the performance of UNet and PUNet is better than MaskRCNN. It means that discriminating between different gas plume objects from each other is not necessary and is even detrimental. As a semantic segmentation method, PUNet reaches the highest IoU of 43.680%, the highest Kappa of 52.992%, and the highest $F1$ score of 43.507% in single-frame segmentation methods. However, MaskRCNN achieves an IoU of 42.405%, a Kappa of 49.542%, and an $F1$ score of 30.446%. The scores of MaskRCNN are lower than UNet or PUNet.

3) Single Frame Versus Multiframe: Adding the consideration of temporal information, spatial–temporal segmentation outperforms single-frame segmentation in general. The proposed CFDNet network reaches the state-of-the-art performance among different methods. Moreover, VSF, TDNNet, and TEANet also have impressive performance on gas detection. As to IoU, CFDNet and TDNNet get the best scores, 50.128% and 48.407%, respectively. As to Kappa, CFDNet and VSF get the best scores, 59.902% and 57.668%, respectively. As to the $F1$ score, CFDNet and PUNet get the best scores, 44.885% and 41.339%, respectively. As to Precision, VSF and CFDNet get the best scores, 50.693% and 48.553%, respectively. As to Recall, TEANet and CFDNet get the best scores, 49.457% and 48.708%, respectively. The single-frame segmentation model PUNet can detect the gas leakage region well in some cases but fails in certain special scenarios, such as the scene in Fig. 12(c) and (d), due to the absence of temporal information. With the temporal motion pattern of gas, the multiframe segmentation approaches can get an approximate prediction of the gas leakage region, which is beneficial for the detection of the next time. Although VSF also considers incorporating physical information into deep learning models, the Gaussian diffusion model it employs is simpler compared to the CFD equations. Therefore, when dealing with small-area gas plumes, VSF's performance is unstable.

In summary, CFDNet achieves the best scores in three metrics and the second-best scores in recall, precision, and metrics. CFDNet has been proven to have the best performance on a real-world dataset. The IoU of the proposed method is 50.128%. The Kappa of the proposed method is 59.902% and the $F1$ score is 44.885%. The proposed method achieved the best performance in comprehensive evaluation metrics.

VI. DISCUSSION

On the basis of qualitative and quantitative evaluation, we conduct ablation studies continually to validate the effectiveness of the proposed LED module and CFD Basic Block modules. Moreover, the sensitivity of detection results to the gas plume area and target signal-to-noise ratio is analyzed. Finally, the complexity of the proposed model is compared with other models.

A. Ablation Studies

To evaluate the proposed LED data enhancement module and the spatial–temporal feature extraction module, CFD Basic Block, we add these modules into the basic backbone, ResNet18, sequentially. The LED data enhancement module is placed in front of the spatial–temporal encoder. Because the ResNet18 performs a total of four layers of basic blocks and downsampling operations during feature extraction, CFD Basic Blocks are inserted into four stages one-by-one. The results of ablation studies are shown in Table IV.

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TABLE IV
ABLATION OF THE PROPOSED MODULES

LED	CFD Basic Block				IoU(%)↑	Kappa(%)↑	F1Score(%)↑
	Stage 1	Stage 2	Stage 3	Stage 4			
×	×	×	×	×	9.833	15.788	16.175
✓	×	×	×	×	23.980	32.968	29.777
✓	✓	×	×	×	28.974	38.186	31.947
✓	✓	✓	×	×	30.020	38.917	30.402
✓	✓	✓	✓	×	35.191	44.479	36.315
✓	✓	✓	✓	✓	50.128	59.902	44.885

1) Ablation of LED: Our backbone is ResNet18, which is a convolutional neural network for feature extraction. Without any innovative module, the backbone fails to detect the mask of the gas leakage region. Three main metrics of the backbone are all below 20%. With the assistance of local entropy and combination differentiation, the feature of gas is enhanced significantly. The model reached an IoU of 23.980%, a Kappa of 32.968%, and an $F1$ score of 29.777%. The capability of the proposed module is partly due to the initial design of the multispectral instrument, which aimed to enhance the gas signals through interband differencing. Furthermore, the local entropy of thermal infrared images markedly enhanced the characteristics of the gas signals.

2) Ablation of CFD Basic Block: The proposed CFD Basic Block was inserted into the four stages of the spatial–temporal encoder. Without the assistance of CFD Basic Block, the combination of backbone and LED data enhancement module reaches an IoU score of 23.980%, a Kappa score of 32.968%, and an $F1$ score of 29.777%. From shallow feature processing to deep feature processing, the four CFD Basic Blocks are sequentially added. All scores significantly improve as the number of modules increases. Further analyzing the impact of the placement of CFD Basic Blocks on the model's performance, we found that placing the CFD Basic Block module in the high-dimensional feature processing part can significantly improve the model's performance. Applying the motion pattern of fluid on the deep feature is beneficial to extract the leakage region of gas.

B. Sensitivity Analysis

1) Sensitivity Analysis of Area Ratio: As shown in Fig. 13, the area ratio of gaseous targets to the camera scene is relatively small [42]. The larger the area ratio is, the better deep learning performs on inference. To be exact, six spatial–temporal encoders all do well when the area ratio of gas varies around 3.5%. Although some methods outperform our proposed method, CFDNet, in specific scenarios, CFDNet demonstrates the most stable performance across various area ratios. This means that CFDNet consistently ranks among the top performers under different gas area ratios. Specifically, under the lowest gas area ratio conditions, the proposed method achieves the best IoU value compared to other methods. It is convincing to prove that the proposed method is more suitable for extracting gas leak areas under low area ratio conditions.

2) Sensitivity Analysis of Target Signal-to-Noise Ratio: Target signal-to-noise ratio measures the saliency of the target

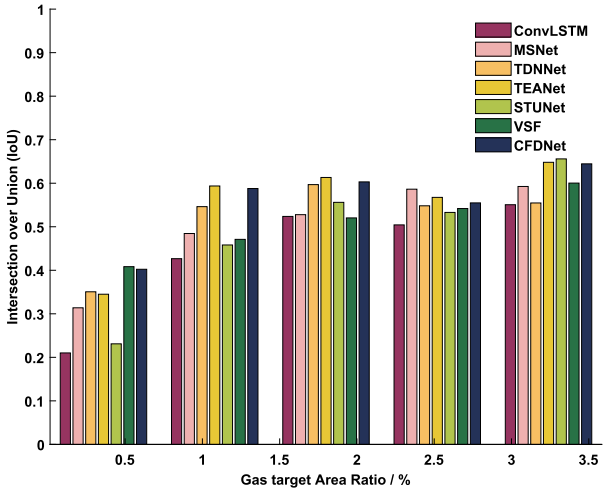


Fig. 13. Sensitivity analysis of area ratio. This graph records the IoU of six spatial–temporal encoders under different area ratios.

on collected images, and its definition can be expressed as follows [25]:

$$\mathrm{SNR}_{\mathrm{target}} = 10 \times \log_{10} \frac{E_{\mathrm{target}} - E_{\mathrm{back}}}{\sigma_{\mathrm{back}}} \tag{26}$$

where E_{target} denotes the mean gray value of the gaseous target region and E_{back} denotes the mean gray value of the background region. σ_{back} is the standard deviation of the gray value in the background region.

As shown in Fig. 14, dividing the signal-to-noise ratio into six classes, we can find that the signal-to-noise ratio of gas in thermal infrared multispectral images is lower than 0 dB in most cases. When the signal-to-noise ratio is less than 0, it indicates that the target signal has a diminishing effect relative to the background signal. The lower the target signal-to-noise ratio, the lower the gas target value relative to the background signal. Therefore, we can find that the performance of various methods is generally poor around 0 dB, but the best performance is achieved around −20 dB. The proposed method, CFDNet, achieved the most stable performance across all different target signal-to-noise ratio levels. The greater the absolute value of the signal-to-noise ratio, the better the model performs, just like the conclusion in [26].

From the above analysis, we can observe that the proposed method exhibits relatively the most stable performance for gas signals under different area ratios and signal-to-noise ratio

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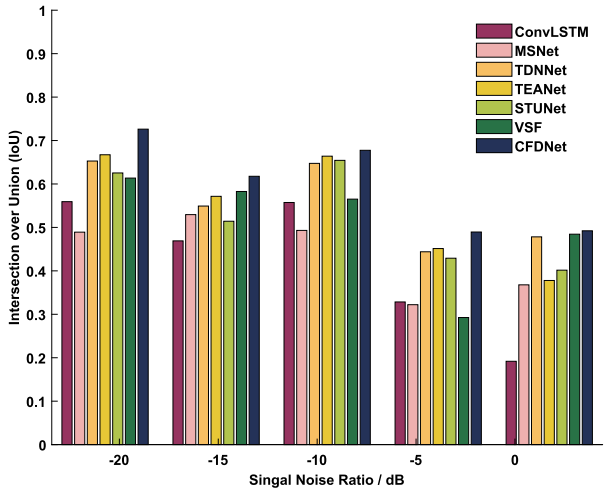


Fig. 14. Sensitivity analysis of signal-to-noise ratio. This graph records the IoU of six spatial–temporal encoders under different target signal-to-noise ratios.

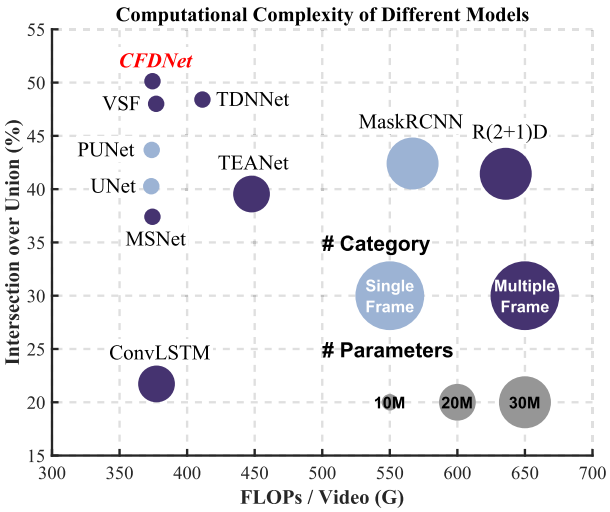


Fig. 15. Computational complexity of different deep learning models. These models were separated into two types, namely, single-frame segmentation and multiframe segmentation. Moreover, the size of the circle denotes the parameters of the model, which was classified into three levels (10, 20, and 30 M).

levels and achieves the best IoU score when the gas area ratio or the absolute value of the signal-to-noise ratio is low.

C. Computational Complexity

To further evaluate the parameter count and computational complexity of different gas detection deep learning models, we recorded the number of parameters and the flops (floating point operations per second) under the same input size conditions. The computational complexities of different single-frame and multiframe gas detection deep learning methods are recorded in Table V and Fig. 15.

According to Fig. 15, adding temporal information into gas detection models increases the volume of models. Therefore, multiframe segmentation models are larger than single-frame segmentation models in general. Specifically, due to the presence of the feature pyramid networks (FPNs) and region proposal network (RPN) modules, Mask R-CNN has a larger size than most single-frame image segmentation methods and even surpasses many multiframe image segmentation methods. Moreover, we can note that R(2+1)D has more parameters than other models. The proposed method, CFDNet, has a small number of parameters. Although CFDNet is a multiframe image segmentation method that incorporates temporal information, its model size is similar to that of single-frame segmentation. It is important to point out that the number of MSNet parameters is fewer than CFDNet. However, the performance of the MSNet model is far inferior to that of the proposed method, CFDNet.

TABLE V
FLOPS AND PARAMETERS OF MODELS

Types	Methods	Backbone	flops(G)↓	params(M)↓
Single Frame	UNet	Resnet18	373.472	14.423
	PUNet	Resnet18	373.805	14.685
	MaskRCNN	Resnet18	566.853	31.002
Multie Frame	ConvLSTM	Resnet18	377.364	25.336
	MSNet	Resnet18	374.332	14.444
	TDNNet	Resnet18	411.333	14.922
	TEANet	Resnet18	447.666	24.408
	R(2+1)D	Resnet18	635.804	36.537
	VSF	Resnet18	377.088	15.439
	proposed	Resnet18	374.361	14.527

VII. CONCLUSION AND FUTURE WORK

Addressing the challenges of weak gas signals and the arbitrary variability of plume shapes in the field of gas detection, we proposed a convolutional neural network based on CFDs, namely, CFDNet, to detect the leakage region of gas. In this network, we innovatively introduced an LED data feature enhancement module combining combinatorial differencing and local entropy. Additionally, based on the Navier–Stokes equations in CFDs, we estimated the velocity field and diffusion coefficient field using convolutional modules. Predicting the displacement and diffusion of feature values in the high-dimensional feature space is beneficial for gas leak region detection through the semantic segmentation decoder. On the other hand, the existing datasets are insufficient to support the training of deep learning models and lack accurate and reliable mask annotations. To solve the problem, we constructed a simulated thermal infrared multispectral video dataset with the assistance of WRF-LES. To further improve the model's generalization on real-world data, we proposed a transfer learning framework with IOD as a pretext task.

The relevant experiments showed that the proposed method performed well in gas detection on real-world datasets. It achieved the best scores in three out of five key metrics, IoU, Kappa, and $F1$ score, and the second-best scores in precision and recall metrics. Overall, CFDNet achieved the best performance in detecting gas leakage regions on the current dataset, with the IoU index of 50.128%, Kappa index of 59.902%, $F1$ score of 44.885%, precision of 48.553%, and recall of 48.708%. Moreover, a more intuitive visual demonstration was provided, specifically explaining that the proposed method achieved good detection results in tackling strong and

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salient leakage, complex gas plume shapes, and small-area gas plumes. The effectiveness of proposed modules was further validated by ablation studies. With the addition of the LED module and CFD Basic Block, the gas detection accuracy of the model was significantly improved. Furthermore, we have analyzed the effect of area ratio and signal-to-noise ratio of gaseous targets on the detection accuracy, IoU. The detection performance of various methods improves as the area ratio and the absolute value of signal-to-noise ratio increase. The proposed method demonstrated the most robust IoU metric under different conditions. Furthermore, it showed a significant advantage under the most extreme conditions of small-area ratios and low absolute signal-to-noise ratios. Finally, the flops and number of parameters of different deep learning models were assessed systematically. As an excellent multiframe temporal segmentation method, the CFDNet model's size is similar to that of a single-frame semantic segmentation method without temporal information. This supports rapid response in gas detection applications.

Although the proposed model works well on given scenarios, the diversity of the dataset is insufficient to build a general gas detector to adapt to different weather conditions and multiple gas sources. In the future, we will focus on building a larger real-world gas leak scenario dataset under diverse weather and gas emission sources. More attention will be put on domain adaptation strategies to reduce the data distribution gap between simulated and real-world data. Based on the dataset, it is important to point out that the CFD Basic Block proposed in this article is a general spatial–temporal feature encoder. It is promising to carry on the following studies based on gas detection with thermal infrared video imaging.

1) *Quantitative Inversion of Gas Density*: The quantitative inversion of gas column concentration is treated as a regression task and is performed using a deep learning model [86]. The regression task is more advanced compared to the segmentation task, as it allows for more accurate extraction of gas leakage information.
2) *The Reconstruction of Fluid Field*: Due to the limitation of camera sampling rate, we can only get the information of gas emission with an FPS lower than 1. Utilizing the temporal motion pattern of fluid, recovering the leakage region or density field of gas with finer temporal resolution is challenging and significant for emergency event assessment [87].
3) *The Prediction of Fluid Field*: With historical thermal infrared multispectral frames, the status of gas is relatively easy to obtain. However, in certain application scenarios, predicting the future diffusion state of chemical gases is important for making decisions [88]. With the proposed spatial–temporal feature extractor, it is feasible to predict the future gas diffusion trend by learning from the historical gas movement trajectories.

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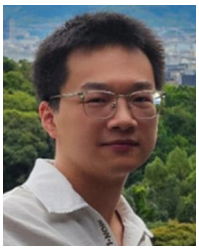
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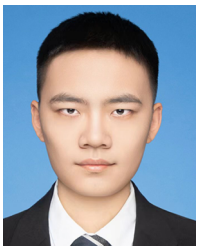
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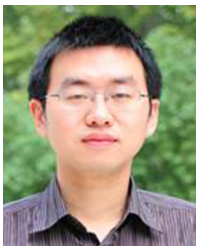
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